

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: THEORY OF STATISTICS

Session: 2025-26 Semester: II

Solutions for Problem Set 3

(Ancillary Statistics, Basu's Theorem, CRLB)

Solution 1: Uniform Distribution $U(\theta, \theta + 1)$:

The population pdf is:

$$f(x) = 1, \quad \theta \leq x \leq \theta + 1$$

The population cdf is:

$$F(x) = x - \theta, \quad \theta \leq x \leq \theta + 1$$

Derivation of the pdf of $X_{(1)}$: The cdf of the first order statistic is:

$$F_{X_{(1)}}(u) = P(X_{(1)} \leq u) = 1 - P(X_{(1)} > u) = 1 - [1 - F(u)]^n$$

Differentiating with respect to u :

$$f_{X_{(1)}}(u) = n[1 - F(u)]^{n-1}f(u)$$

$$f_{X_{(1)}}(u) = n[1 - (u - \theta)]^{n-1}(1)$$

$$f_{X_{(1)}}(u) = n(\theta + 1 - u)^{n-1}, \quad \theta \leq u \leq \theta + 1$$

Derivation of the pdf of $X_{(n)}$: The cdf of the n -th order statistic is:

$$F_{X_{(n)}}(v) = P(X_{(n)} \leq v) = [F(v)]^n$$

Differentiating with respect to v :

$$f_{X_{(n)}}(v) = n[F(v)]^{n-1}f(v)$$

$$f_{X_{(n)}}(v) = n(v - \theta)^{n-1}(1)$$

$$f_{X_{(n)}}(v) = n(v - \theta)^{n-1}, \quad \theta \leq v \leq \theta + 1$$

Derivation of the Joint pdf of $(X_{(1)}, X_{(n)})$: The joint density for $\theta \leq u < v \leq \theta + 1$ is given by:

$$f_{X_{(1)}, X_{(n)}}(u, v) = \frac{n!}{(1-1)!(n-1-1)!(n-n)!} [F(u)]^0 [F(v) - F(u)]^{n-2} [1 - F(v)]^0 f(u)f(v)$$

$$f_{X_{(1)}, X_{(n)}}(u, v) = n(n-1)[(v - \theta) - (u - \theta)]^{n-2}(1)(1)$$

$$f_{X_{(1)}, X_{(n)}}(u, v) = n(n-1)(v - u)^{n-2}$$

Distribution of the Range $R = X_{(n)} - X_{(1)}$: Let $R = X_{(n)} - X_{(1)}$ and $M = X_{(1)}$. The Jacobian $J = 1$.

$$f_{R,M}(r, m) = n(n-1)r^{n-2}, \quad 0 \leq r \leq 1, \quad \theta \leq m \leq \theta + 1 - r$$

Integrating out M :

$$f_R(r) = \int_{\theta}^{\theta+1-r} n(n-1)r^{n-2} dm = n(n-1)r^{n-2}(1-r), \quad 0 \leq r \leq 1$$

Since $f_R(r)$ is free of θ , R is **ancillary**.

Solution 2: Bernoulli Distribution:**Fisher Information Derivation:**

$$\ln f(x; p) = x \ln p + (1 - x) \ln(1 - p)$$

$$\frac{\partial^2}{\partial p^2} \ln f = -\frac{x}{p^2} - \frac{1-x}{(1-p)^2} \implies I(p) = E \left[\sum_{i=1}^n \left(\frac{X_i}{p^2} + \frac{1-X_i}{(1-p)^2} \right) \right] = \frac{n}{p(1-p)}$$

CRLB:

$$\text{CRLB} = \frac{[g'(p)]^2}{I(p)} = \frac{1^2}{\frac{n}{p(1-p)}} = \frac{p(1-p)}{n}$$

Solution 3: Normal Distribution $N(\mu, \sigma^2)$:**Exponential Form:**

$$f(x; \mu) = \left[\frac{1}{\sqrt{2\pi}\sigma^2} e^{-x^2/2\sigma^2} \right] \left[e^{-\mu^2/2\sigma^2} \right] \exp \left(\frac{\mu}{\sigma^2} \cdot x \right)$$

Identifying $w(\mu) = \mu/\sigma^2$, we confirm $\bar{X}$ is complete sufficient. By Basu's Theorem, $\bar{X}$ and S^2 are independent.

Solution 4: Normal Distribution $N(\theta, 1)$:

Derivation of Fisher Information $I(\theta)$: The joint log-likelihood for the sample $X_1, \dots, X_n$ is:

$$\ln L(\theta) = \sum_{i=1}^n \ln f(x_i; \theta) = \sum_{i=1}^n \left(-\frac{1}{2} \ln(2\pi) - \frac{(x_i - \theta)^2}{2} \right)$$

Taking the first derivative with respect to θ :

$$\frac{\partial}{\partial \theta} \ln L(\theta) = \sum_{i=1}^n (x_i - \theta)$$

Taking the second derivative:

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \sum_{i=1}^n (-1) = -n$$

The Fisher Information is defined as:

$$I(\theta) = -E \left[\frac{\partial^2}{\partial \theta^2} \ln L(\theta) \right] = -E[-n] = n$$

CRLB for $g(\theta) = \theta^2$: We have $g(\theta) = \theta^2 \implies g'(\theta) = 2\theta$. Following the formula :

$$\text{CRLB} = \frac{[g'(\theta)]^2}{I(\theta)}$$

Substituting our derived values:

$$\text{CRLB} = \frac{(2\theta)^2}{n} = \frac{4\theta^2}{n}$$

CRLB for $g(\theta) = \theta$: Here $g'(\theta) = 1$.

$$\text{CRLB} = \frac{1^2}{n} = \frac{1}{n}$$

Since $\text{Var}(\bar{X}) = 1/n$, the sample mean $\bar{X}$ attains the bound.

Variance of UMVUE $\delta = \bar{X}^2 - \frac{1}{n}$:

$$\text{Var}(\delta) = \frac{4\theta^2}{n} + \frac{2}{n^2}$$

Since $2/n^2 > 0$, the CRLB for θ^2 is not attained.

Solution 5: Exponential Distribution $\text{Exp}(\lambda)$:

Fisher Information for $\theta = 1/\lambda$: Joint log-likelihood: $\ln L(\theta) = -n \ln \theta - \frac{\sum x_i}{\theta}$.

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{n}{\theta^2} - \frac{2 \sum x_i}{\theta^3} \implies I(\theta) = -E \left[\frac{n}{\theta^2} - \frac{2 \sum X_i}{\theta^3} \right] = \frac{n}{\theta^2}$$

CRLB:

$$\text{CRLB} = \frac{1^2}{I(\theta)} = \frac{\theta^2}{n}$$

Appendix: Summary Table

Distribution	Complete Sufficient	Fisher Info $I(\theta)$	CRLB (for θ)
Bernoulli (p)	$\sum X_i$	$\frac{n}{p(1-p)}$	$\frac{p(1-p)}{n}$
Poisson (λ)	$\sum X_i$	$\frac{n}{\lambda}$	$\frac{\lambda}{n}$
Normal ($\mu, 1$)	$\sum X_i$	n	$\frac{1}{n}$
Exp (λ)	$\sum X_i$	$\frac{n}{\lambda^2}$	$\frac{\lambda^2}{n}$